Bhavik Dhabaliya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Disha Delphi Public School, Kota	Class XII (CBSE)	2024	89.20%
B V Bhavans Llyods Vidya Niketan, Wardha	Class X (CBSE)	2022	98.20%

SKILLS

Languages: Python, C++, C, SQL

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn

Tools & Technologies: Git, Github, Docker, LaTeX, VS Code, Jupyter Notebook.

Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Operating Systems (OS)

Soft Skills: Leadership, Teamwork, Communication, Problem Solving

PROJECTS

Concurrency Benchmark -- C++20, Multithreading, Atomics, Pandas | GitHub Link

- Implemented three **C++20 spinlocks** from scratch (**TAS**, **CAS**, **bounded-CAS** with starvation-freedom) behind a uniform interface, benchmarked against std::mutex on a **multithreaded Sudoku validator** via a compile-time-templated validator.
- Built a benchmark harness (lock × thread count × chunk × grid size) with CI correctness tests, then analyzed results in pandas/matplotlib to surface the parallel/sequential crossover and core-scaling behavior.

Hospital Readmission Prediction -- Python, scikit-learn, XGBoost | GitHub Link

- Built a **leak-safe scikit-learn pipeline** (~100K records, 9% positive class) and benchmarked a five-model ladder(**Logistic Regression -> Decision Tree** -> **Random forest** -> **XGBoost**) on **PR-AUC** with out-of-fold threshold selection.
- Caught a probability-calibration artifact distorting the comparison metric and re-measured all models; used SHAP to show additive, bagged, and boosted models converge at **PR-AUC 0.17**, concluding the ceiling was feature information rather than model capacity.

API Gateway - Python, FastAPI, Redis, Docker, Prometheus/Grafana | GitHub Link

- Built a reverse-proxy **API gateway** (Python/FastAPI, Redis, Docker) featuring **atomic Redis-Lua rate limiting** (token bucket & sliding window), TTL response caching, and Prometheus/Grafana observability, deployable in one command.
- Load-tested with k6 at **~595 req/s** across **53,000+ requests** with **100% correctness**; verified resilience by shedding **~38% of burst traffic** as controlled 429s.

1D Advection PDE Solver -- Python, NumPy, Matplotlib | GitHub Link

- Built a **Lax-Friedrichs PDE solver** in **Python/NumPy** for a coupled 1D hyperbolic system with fully parameterized grid resolution, validated against an exact analytical solution.
- Quantified numerical accuracy through a grid-refinement convergence study and demonstrated the CFL stability limit across Courant-number regimes.

RELEVANT COURSES

Computer Science: Data Structures and Applications, Computational Physics, Introduction to Artificial Intelligence, Introduction to Programming.

Mathematics: Linear Algebra, Statistics, Advanced Mathematics for Physics, Mathematics for Physics, Differential Equations, Calculus I, Calculus II.

Physics: Electricity & Magnetism, Special Relativity, Classical Mechanics, Thermodynamics, Optics & Photonics, Fluid Dynamics.

POSITIONS OF RESPONSIBILITY

Operations Manager, E-Cell IIT Hyderabad

- Led logistics for the E-Cell's entrepreneurship events from start to finish, handling the schedule, the budget, the vendors, and the team so everything came together on the day.

Public Relations & Outreach Core, Tinkerers' Lab, IIT Hyderabad

- Managed PR and outreach initiatives to boost lab visibility and engagement across campus and external audiences.

Core Member, Rang De Manch (Drama Club), IIT Hyderabad

- Contributed to script development and rehearsals as part of the club's core team, organizing performances and participated in Inter-College street plays.

EXTRACURRICULAR

- Solved **350+ DSA** and **50+ SQL** problems on **LeetCode**, strengthening algorithmic thinking and database querying skills
- **Theatre Team, Inter IIT Cultural Meet 7.0 & 8.0** | Represented IIT Hyderabad in Street Play and Stage Play under the Theatre Cup.
- Volunteered with **National Service Scheme (NSS)** at **IIT Hyderabad,** promoting **community service** through **social outreach.**